PHASE 9 · OS & INFRASTRUCTURE SECURITY

OS & Infrastructure Hardening



Linux Security & CIS Benchmarks

CIS Benchmarks | Kernel Hardening | SELinux/AppArmor | Seccomp | Rootkit Detection

Time estimate: 5–6 hours

CHAPTER GOAL

Harden the OS & Infrastructure L

Implement CIS Benchmarks, kernel hardening, SELinux/AppArmor, syscall filtering, and container escape prevention across the Linux and Kubernetes stack.

Estimated time

5–6 hours

Ubuntu 22.04 · Kubernetes · AppArmor · auditd

By the end of this chapter

- CIS Benchmarks applied (Ubuntu 22.04)
- Kernel hardening parameters
- SELinux/AppArmor enforced
- Syscall filtering (seccomp)
- Container escape prevention
- Rootkit detection
- Compliance with security standards

What is CIS?

The Center for Internet Security publishes consensus-driven configuration benchmarks that define a vetted security baseline for operating systems and infrastructure.



Ubuntu 22.04

Benchmark target

Best practices
for OS hardening

Consensus-driven
recommendations

Evaluated
against major threats

Compliance framework
PCI-DSS, HIPAA, SOC2

Improves
security baseline

CIS Ubuntu 22.04 Checklist

Initial Setup

- Disable unnecessary filesystems
- Configure /tmp, /var/tmp, /var/log permissions
- Bind mount /var/tmp to /tmp

Access Control

- World-writable files audit
- SUID/SGID binaries audit
- Proper umask settings (0077)

System Accounting

- auditd enabled
- Audit rules for system calls
- Centralized log collection

Sysctl Hardening Parameters

/etc/sysctl.d/99-hardening.conf

```
kernel.kptr_restrict = 2
kernel.dmesg_restrict = 1
kernel.unprivileged_bpf_disabled = 1
kernel.unprivileged_userns_clone = 0

kernel.randomize_va_space = 2

kernel.panic_on_oops = 1
kernel.panic = 60

net.ipv4.tcp_syncookies = 1
net.ipv4.conf.all.rp_filter = 1

sysctl -p /etc/sysctl.d/99-hardening.conf
```

kptr_restrict / dmesg_restrict

Hide kernel pointers & dmesg from unprivileged users

unprivileged_bpf_disabled

Blocks BPF-based privilege escalation paths

randomize_va_space = 2

Full ASLR — randomizes memory layout

tcp_syncookies / rp_filter

Mitigates SYN floods & IP spoofing

Seccomp & Container Hardening

Block dangerous syscalls at the Kubernetes pod level to reduce the container's kernel attack surface.

hardened-pod.yaml

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: hardened-pod
spec:
  securityContext:
    seccompProfile:
      type: RuntimeDefault
  containers:
  - name: app
    image: myapp:latest
    securityContext:
      allowPrivilegeEscalation: false
      readOnlyRootFilesystem: true
      runAsNonRoot: true
      runAsUser: 1000
      capabilities:
        drop:
        - ALL
```

`seccompProfile: RuntimeDefault`

Applies the container runtime's default syscall filter

`allowPrivilegeEscalation: false`

Prevents processes from gaining more privileges than their parent

`readOnlyRootFilesystem: true`

Root filesystem cannot be modified at runtime

`capabilities.drop: ALL`

Strips all Linux capabilities by default — least privilege

AppArmor Profiles

Confine what a process can access — files, network, capabilities, and mounts — via a mandatory access control profile.

/etc/apparmor.d/containers/kubernetes

```
aa-status

#include <tunables/global>

profile kubernetes-default flags=(attach_disconnected) {
  #include <abstractions/base>
  #include <abstractions/nameservice>

  network,
  capability,
  file,
  mount,
}

apparmor_parser -r /etc/apparmor.d/containers/kubernetes
```

1 **Check status**

aa-status shows loaded profiles & modes

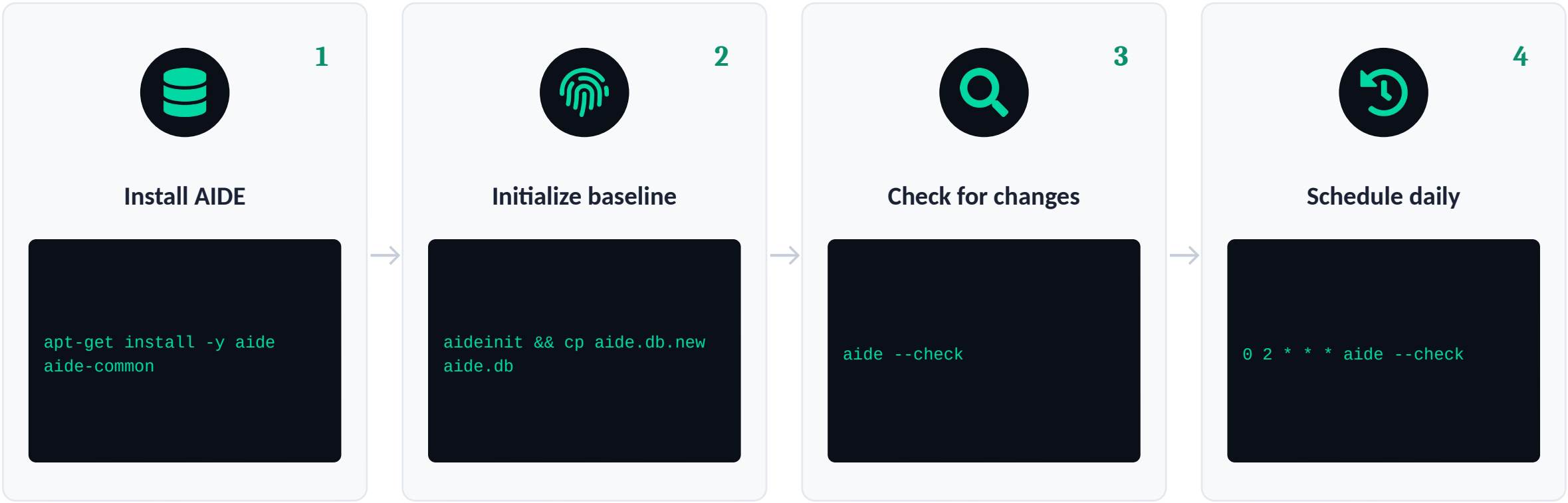
2 **Define scope**

Grant only required abstractions

3 **Load profile**

apparmor_parser -r reloads the profile

Rootkit Detection with AIDE



AIDE detects unauthorized file changes — a core signal for rootkit and intrusion detection.

auditd — Forensics & Compliance

bash

```
systemctl enable auditd

auditctl -w /etc/passwd -p wa -k passwd_changes
auditctl -w /etc/group  -p wa -k group_changes
auditctl -w /etc/sudoers -p wa -k sudoers_changes

ausearch -k passwd_changes
```

/etc/passwd

`-k passwd_changes`

User account modifications

/etc/group

`-k group_changes`

Group membership changes

/etc/sudoers

`-k sudoers_changes`

Privilege escalation rules

29.7 CHAPTER SUMMARY

Defense in Depth at the OS Level

CONTROL	PURPOSE
CIS Benchmarks	Security baseline
Kernel Hardening	Prevent exploits
Seccomp	Block dangerous syscalls
AppArmor	Mandatory access control
File Integrity	Rootkit detection
Audit Logging	Forensics & compliance



KEY TAKEAWAY

Defense in depth at the OS level prevents container escapes and privilege escalation.

CIS Benchmarks | Kernel Params | Seccomp | AppArmor | AIDE | auditd